Solid Waste Management and Decentralized Composting Audit 2024

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SOLID WASTE MANAGEMENT AND DECENTRALIZED COMPOSTING AUDIT 2024

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Authoring Body: Campus Environmental Sanitation Committee and Nature Club.

Audited Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. QUANTITATIVE WASTE GENERATION PROFILE

An intensive institutional waste composition and mass-balance audit was conducted across the 32,500 m2 campus of Vivekananda Education Society's Institute of Technology (VESIT Green Campus) during academic year 2024. Serving a campus community of 3,200 individuals, the institution generated a total of 68,400 kg (68.40 metric tonnes) of solid municipal and electronic waste.

Characterization across 12 sampling cycles established the following waste composition breakdown:

- Organic and Kitchen Waste: 42.0% (28,728 kg/year), generated primarily by Canteen dining (BLD_CAN), food preparation peelings, leftover student meals, and botanical landscape trimmings.

- Recyclable Paper, Cardboard, and Stationeries: 33.0% (22,572 kg/year), originating from examination booklets, administrative archives, packaging crates, and Library (BLD_LIB) periodicals.

- Post-Consumer Plastics and Multi-Layer Packaging: 14.0% (9,576 kg/year), comprising beverage containers, snack wrappers, and packaging films.

- Electronic Waste (E-Waste) and Laboratory Scrap: 3.0% (2,052 kg/year), consisting of obsolete motherboard PCBs, power supplies, CRT/TFT monitors, lab wiring, and spent batteries from Computer Laboratories (BLD_LAB).

- Non-Recyclable Inert and Sanitary Refuse: 8.0% (5,472 kg/year), including dust, sweepings, and sanitary bathroom waste.

2. PERFORMANCE OF DECENTRALIZED ORGANIC TREATMENT (INIT_BIOGAS_CANTEEN)

To address the primary organic waste fraction, VESIT commissioned project INIT_BIOGAS_CANTEEN behind the Canteen building. The installation comprises:

- An anaerobic biomethanation plant designed with a rated processing capacity of 250 kg of organic waste per day.

- Twin 500-litre dual-axis aerobic composting tumblers with mechanical aeration for horticultural dry leaves and cellulosic paper napkins.

In calendar year 2024, the biomethanation digester successfully ingested and processed 19,200 kg of canteen wet food scraps. The anaerobic digestion process generated an average of 12 m3 of methane-rich biogas daily. This fuel is piped via stainless steel conduits directly into the commercial Canteen kitchen to supply two heavy-duty cooking burners, displacing 14 commercial liquefied petroleum gas (LPG) cylinders (19 kg each) per month.

Environmental and carbon accounting confirms that diverting 19,200 kg of putrescible organic waste away from the municipal Deonar open landfill prevented uncontrolled anaerobic decomposition and fugitive methane venting. Utilizing emission factor EF_WASTE_LANDFILL (1.25 kg CO2e/kg waste) versus composting factor EF_WASTE_COMPOST (0.08 kg CO2e/kg waste), the composting and biogas plant achieved a net carbon abatement of 22,464 kg CO2e in 2024. Furthermore, the digester generated 1,400 kg of nutrient-dense organic bioslurry and bio-compost, utilized across the 600 m2 Miyawaki native forest (INIT_BIODIVERSITY_ZONE) and campus botanical beds, eliminating synthetic chemical fertilizers.

3. DRY RECYCLABLES RECOVERY AND AUTHORIZED E-WASTE PARTNERSHIPS

Under the campus Circular Economy protocol, segregated dry paper and cardboard (22,572 kg) were sorted in the central sorting yard. Through formal tie-ups with municipal recyclers, 19,800 kg of post-consumer paper was recovered for pulping. Applying avoided paper production factor EF_WASTE_RECYCLE (-0.65 kg CO2e/kg waste), paper recovery yielded an avoided carbon credit of -12,870 kg CO2e.

Electronic waste disposal is governed by the national E-Waste Management Rules 2022. VESIT entered into a formal memorandum with authorized e-waste recycler Greenscape Solutions. During 2024, 1,850 kg of decommissioned computer

electronics from BLD_LAB and DEP_CMPN underwent certified downstream material extraction, recovering precious copper, gold, aluminum, and rare earths while preventing hazardous heavy metal contamination (cadmium, lead) of Mumbai groundwaters, directly supporting United Nations SDG 12 (Responsible Consumption and Production).

4. PROGRESS TOWARD TARGET TRG_ZERO_WASTE

In 2024, the institution attained an aggregate solid waste diversion rate of 48.2% (composting + formal recycling). This represents measurable progress toward target TRG_ZERO_WASTE, which mandates achieving an 85.0% diversion rate and zero organic waste to landfill by December 31, 2028. Key measures for 2025 include expanding student cafeteria plate-waste audits and enforcing a total prohibition on single-use plastics.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_WST_006**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*